

Hints & Solutions

#MathBoleTohMathonGo

Q1 (1) - Single Correct

Consider

$$\lim_{n \rightarrow \infty} \frac{1^2 \cdot n + 2^2 \cdot (n-1) + 3^2 \cdot (n-2) + \dots + n^2 \cdot 1}{1^3 + 2^3 + 3^3 + \dots + n^3}$$

$$= \lim_{n \rightarrow \infty} \frac{1^2 \cdot n + 2^2 \cdot (n-1) + 3^2 \cdot (n-2) + \dots + n^2 \{n - (n-1)\}}{\Sigma n^3} \dots (i)$$

Now, consider Numerator

$$= n(1^2 + 2^2 + 3^2 + \dots + n^2) - \{1 \cdot 2^2 + 2 \cdot 3^2 + \dots + (n-1) \cdot n^2\}$$

$$= n \Sigma n^2 - \sum_{r=2}^n (r-1)r^2$$

$$= n \Sigma n^2 - \sum_{r=2}^n (r^3 - r^2) = n \Sigma n^2 - \sum_{r=1}^n (r^3 - r^2)$$

$$= n \Sigma n^2 - \sum_{r=1}^n r^3 + \sum_{r=1}^n r^2 = n \Sigma n^2 - \Sigma n^3 + \Sigma n^2$$

$$= (n+1) \Sigma n^2 + \Sigma n^3$$

Now from Eq. (i), we get

$$\lim_{n \rightarrow \infty} \frac{1^2 \cdot n + 2^2 \cdot (n-1) + 3^2 \cdot (n-2) + \dots + n^2 \cdot 1}{1^3 + 2^3 + 3^3 + \dots + n^3}$$

$$= \lim_{n \rightarrow \infty} \frac{[(n+1) \Sigma n^2 - \Sigma n^3]}{\Sigma n^3} = \lim_{n \rightarrow \infty} \frac{(n+1) \Sigma n^2}{\Sigma n^3} - 1$$

$$= \lim_{n \rightarrow \infty} \frac{(n+1)n(n+1)(2n+1)}{6n^2(n+1)^2} \times 4 - 1$$

$$\left[\because \Sigma n^2 = \frac{n(n+1)(2n+1)}{6}, \Sigma n^3 = \frac{n^2(n+1)^2}{4} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{2(2n+1)}{3n} - 1$$

$$= \frac{2}{3} \left[\lim_{n \rightarrow \infty} \left(2 + \frac{1}{n} \right) \right] - 1$$

$$= \frac{2}{3} \times 2 - 1 = \frac{4}{3} - 1 = \frac{1}{3}$$

Q2 (1) - Single Correct

Consider

$$\lim_{x \rightarrow \infty} \frac{\cot^{-1}(x^{-a} \log_a x)}{\sec^{-1}(a^x \log_e a)} = \lim_{x \rightarrow \infty} \frac{\cot^{-1}\left(\frac{\log_a x}{x^a}\right)}{\sec^{-1}\left(\frac{a^x}{\log_a x}\right)} \dots (i)$$

Note that $\lim_{x \rightarrow \infty} \left(\frac{\log_a x}{x^a} \right) = \lim_{x \rightarrow \infty} \left(\frac{\frac{1}{x} \times \log_a e}{a x^{a-1}} \right)$ [using L' Hospital's rule]

$$= \lim_{x \rightarrow \infty} \left(\frac{\log_a e}{a x^a} \right) = 0 \quad [\because a > 1 \text{ and } x \rightarrow \infty]$$

and $\lim_{x \rightarrow \infty} \left(\frac{a^x}{\log_a x} \right) = \lim_{x \rightarrow \infty} \left(\frac{a^x \log_e a}{\log_a e \cdot \frac{1}{x}} \right)$ [using L' Hospital's rule]

$$= \lim_{x \rightarrow \infty} \left(\frac{x \cdot a^x \log_e a}{\log_a e} \right)$$

$$= \infty$$

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From Eq. (i), we get

$$\lim_{x \rightarrow \infty} \frac{\cot^{-1}(x^{-a} \log_a x)}{\sec^{-1}(a^x \log_x a)} = \frac{\cot^{-1}(0)}{\sec^{-1}(\infty)} = \frac{\frac{\pi}{2}}{\frac{\pi}{2}} = 1$$

Q3 (1) - Single Correct

By using graph, we can see that

$$\text{when } x \rightarrow \frac{\pi}{4}^+, \sec\left(x - \frac{\pi}{4}\right) \rightarrow 1^+ \Rightarrow \left[\sec\left(x - \frac{\pi}{4}\right)\right] = 1$$

$$\operatorname{cosec}\left(x + \frac{\pi}{4}\right) \rightarrow 1^+ \Rightarrow \left[\operatorname{cosec}\left(x + \frac{\pi}{4}\right)\right] = 1$$

$$\cot\left(x + \frac{\pi}{4}\right) \rightarrow 0^- \Rightarrow \left[\cot\left(x + \frac{\pi}{4}\right)\right] = -1$$

$$\tan\left(x - \frac{\pi}{4}\right) \rightarrow 0^+ \Rightarrow \left[\tan\left(x - \frac{\pi}{4}\right)\right] = 0$$

$$\text{and when } x \rightarrow \frac{\pi}{4}^-$$

$$\sec\left(x - \frac{\pi}{4}\right) \rightarrow 1^+ \Rightarrow \left[\sec\left(x - \frac{\pi}{4}\right)\right] = 1$$

$$\operatorname{cosec}\left(x + \frac{\pi}{4}\right) \rightarrow 1^+ \Rightarrow \left[\operatorname{cosec}\left(x + \frac{\pi}{4}\right)\right] = 1$$

$$\cot\left(x + \frac{\pi}{4}\right) \rightarrow 0^+ \Rightarrow \left[\cot\left(x + \frac{\pi}{4}\right)\right] = 0$$

$$\tan\left(x - \frac{\pi}{4}\right) \rightarrow 0^- \Rightarrow \left[\tan\left(x - \frac{\pi}{4}\right)\right] = -1$$

$$\text{Thus } \lim_{x \rightarrow \frac{\pi}{4}^+} \left[\sec\left(x - \frac{\pi}{4}\right)\right] = 1$$

$$\lim_{x \rightarrow \frac{\pi}{4}^+} \left[\operatorname{cosec}\left(x + \frac{\pi}{4}\right)\right] = 1$$

$$\text{but } \lim_{x \rightarrow \frac{\pi}{4}^+} \left[\cot\left(x + \frac{\pi}{4}\right)\right]$$

$$\text{and } \lim_{x \rightarrow \frac{\pi}{4}^+} \left[\tan\left(x - \frac{\pi}{4}\right)\right] \text{ does not exist}$$

Hence, I_1, I_2 exist and equal to 0 and $\frac{\pi}{2}$ respectively but I_3 and I_4 does not exist.[since, if $\lim_{x \rightarrow a} g(x)$ does not exist, then $\lim_{x \rightarrow a} f \circ g(x) = f[\lim_{x \rightarrow a} g(x)]$ does not exist]

Q4 (4) - Single Correct

$$\text{Consider, } \lim_{x \rightarrow 0^+} \frac{[\tan(\{x\}-1)] \sin\{x\}}{\{x\}(\{x\}-1)}$$

$$= \lim_{h \rightarrow 0} \frac{[\tan(\{0+h\}-1)] \sin\{0+h\}}{\{0+h\}(\{0+h\}-1)}$$

$$= \lim_{h \rightarrow 0} \frac{[\tan(h-1)] \sin h}{h(h-1)} = \lim_{h \rightarrow 0} \frac{\tan(h-1)}{(h-1)} \times \lim_{h \rightarrow 0} \frac{\sin h}{h}$$

$$= \frac{\tan(-1)}{-1} \times 1 = \tan 1 \dots (i)$$

$$\text{and } \lim_{x \rightarrow 0^-} \frac{[\tan(\{x\}-1)] \sin\{x\}}{\{x\}(\{x\}-1)}$$

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$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \frac{[\tan(\{0-h\}-1)] \sin\{0-h\}}{\{0-h\}(\{0-h\}-1)} \\
 &= \lim_{h \rightarrow 0} \frac{[\tan\{(1-h)-9\}] \sin(1-h)}{(1-h)\{(1-h)-1\}} \quad [\because \{0-h\} = 1-h] \\
 &= \lim_{h \rightarrow 0} \frac{\tan(-h) \sin(1-h)}{(1-h)(-h)} = \lim_{h \rightarrow 0} \frac{\tan h}{h} \times \lim_{h \rightarrow 0} \frac{\sin(1-h)}{(1-h)} \\
 &= 1 \times \frac{\sin 1}{1} = \sin 1 \dots (i)
 \end{aligned}$$

From Eqs. (i) and (ii), we get

$$\lim_{x \rightarrow 0} \frac{[\tan(\{x\}-1)] \sin\{x\}}{\{x\}(\{x\}-1)} \text{ does not exist.}$$

Q5 (2) - Single Correct

We have,

$$\begin{aligned}
 P_n &= \prod_{k=1}^n \left(1 + \frac{2}{k^2 + 3k} \right) \\
 &= \prod_{k=1}^n \left(\frac{k^2 + 3k + 2}{k^2 + 3k} \right) \\
 &= \prod_{k=1}^n \left[\frac{(k+1)(k+2)}{k(k+3)} \right] \\
 &= \left(\frac{2 \cdot 3}{1 \cdot 4} \right) \times \left(\frac{3 \cdot 4}{2 \cdot 5} \right) \times \left(\frac{4 \cdot 5}{3 \cdot 6} \right) \cdots \left[\frac{(n+1)(n+2)}{n(n+3)} \right] \\
 &= \frac{3 \cdot [2 \cdot 3 \cdot 4 \cdots (n+1)]}{(1 \cdot 2 \cdot 3 \cdots n) \cdot (n+3)} \\
 \Rightarrow P_n &= \frac{3(n+1)!}{n!(n+3)} = \frac{3(n+1)}{(n+3)}
 \end{aligned}$$

$$\text{Now, } \lim_{n \rightarrow \infty} P_n = 3 \lim_{n \rightarrow \infty} \frac{(n+1)}{(n+3)} = 3 \lim_{n \rightarrow \infty} \frac{\left(1 + \frac{1}{n}\right)}{\left(1 + \frac{3}{n}\right)} = 3$$

Q6 (2) - Single Correct

We have, $a_n = \sum_{k=1}^n 2k$ and $b_n = \sum_{k=1}^n (2k-1)$, where $n \in N$.

$$\text{Clearly, } a_n = 2 \sum_{k=1}^n k = 2 \frac{n(n+1)}{2} = n^2 + n$$

$$\text{and } b_n = \sum_{k=1}^n (2k) - \sum_{k=1}^n 1 = 2 \frac{n(n+1)}{2} - n$$

$$\text{Now, } \lim_{n \rightarrow \infty} (\sqrt{a_n} - \sqrt{b_n}) = \lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - \sqrt{n^2})$$

$$= \lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n)$$

$$= \lim_{n \rightarrow \infty} \frac{n}{\sqrt{n^2 + n} + n} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{n}} + 1}$$

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[dividing numerator and denominator by n^2]

$$= \frac{1}{2}$$

Q7 (2) - Single Correct

We can see that

$$\text{when } x \rightarrow 0^+, \sin x \rightarrow 0^+ \Rightarrow [\sin x] = 0$$

$$\cos x \rightarrow 1^- \Rightarrow [\cos x] = 0$$

$$\tan x \rightarrow 0^+ \Rightarrow [\tan x] = 0$$

and when $x \rightarrow 0^-$

$$\sin x \rightarrow 0^- \Rightarrow [\sin x] = -1$$

$$\cos x \rightarrow 1^- \Rightarrow [\cos x] = 0$$

$$\tan x \rightarrow 0^- \Rightarrow [\tan x] = -1$$

$$\text{Now, } \lim_{x \rightarrow 0^+} (\sin^{-1}[\sin x] + \cos^{-1}[\cos x] - 2 \tan^{-1}[\tan x])$$

$$= \sin^{-1}(0) + \cos^{-1}(0) - 2 \tan^{-1}(0)$$

$$= 0 + \frac{\pi}{2} - 2(0) = \frac{\pi}{2}$$

$$\text{and } \lim_{x \rightarrow 0^-} (\sin^{-1}[\sin x] + \cos^{-1}[\cos x] - 2 \tan^{-1}[\tan x])$$

$$= \sin^{-1}(-1) + \cos^{-1}(0) - 2 \tan^{-1}(-1)$$

$$= -\sin^{-1}(1) + \cos^{-1}(0) + 2 \tan^{-1}(1)$$

$$= -\frac{\pi}{2} + \frac{\pi}{2} + 2 \cdot \frac{\pi}{4}$$

$$\left[\begin{array}{l} \because \sin^{-1}(-x) = -\sin^{-1}(x) \\ \tan^{-1}(-x) = -\tan^{-1}(x) \end{array} \right]$$

$$= \frac{\pi}{2}$$

$$\tan^{-1}(-x) = -\sin^{-1}(-x) = -\sin^{-1}(x)$$

$$\text{Hence, } \lim_{x \rightarrow 0} (\sin^{-1}[\sin x] + \cos^{-1}[\cos x] - 2 \tan^{-1}[\tan x]) = \frac{\pi}{2}$$

Q8 (4) - Single Correct

$$\text{Consider } \lim_{x \rightarrow 2^+} \frac{[x]}{\{x\}} = \lim_{h \rightarrow 0} \frac{[2+h]}{\{2+h\}}$$

$$= \lim_{h \rightarrow 0} \frac{2}{h}, \text{ which does not exist.}$$

$$\text{Thus, } \lim_{x \rightarrow 2} \frac{[x]}{\{x\}} = 1, \text{ does not exist.}$$

Now, consider

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$$\lim_{x \rightarrow 2^+} \frac{\{x\}}{[x]} = \lim_{h \rightarrow 0} \frac{\{2+h\}}{[2+h]} = \lim_{h \rightarrow 0} \frac{h}{2} = 0$$

$$\text{and } \lim_{x \rightarrow 2^-} \frac{\{x\}}{[x]} = \lim_{h \rightarrow 0} \frac{\{2-h\}}{[2-h]} = \lim_{h \rightarrow 0} \frac{\{1+(1-h)\}}{1} \\ = \lim_{h \rightarrow 0} (1-h) = 1$$

Thus, $\lim_{x \rightarrow 2} \frac{\{x\}}{[x]} = m$, does not exist.

Hence, neither I nor m exists.

Q9 (1) - Single Correct

$$\text{Let } f(n) = \sum_{r=1}^n \frac{r^2}{n^3+n^2+r}$$

$$= \frac{1^2}{n^3+n^2+1} + \frac{2^2}{n^3+n^2+2} + \dots + \frac{n^2}{n^3+n^2+n} \dots (i)$$

Now, consider

$$g(n) = \frac{1^2}{n^3+n^2+n} + \frac{2^2}{n^3+n^2+n} + \dots + \frac{n^2}{n^3+n^2+n} \dots (ii)$$

$$= \frac{(1^2 + 2^2 + \dots + n^2)}{n^3 + n^2 + n} \\ = \frac{\Sigma n^2}{n(n+1)(2n+1)} \\ = \frac{n^3 + n^2 + n}{6(n^3 + n^2 + n)} \\ = \frac{(n+1)(2n+1)}{6(n^2 + n + 1)}$$

$$\text{and } h(n) = \frac{1^2}{n^3+n^2+1} + \frac{2^2}{n^3+n^2+1} + \dots + \frac{n^2}{n^2+n+1} \dots (iv)$$

$$= \frac{(1^2 + 2^2 + \dots + n^2)}{n^3 + n^2 + 1} \\ = \frac{\Sigma n^2}{n^3 + n^2 + 1} = \frac{n(n+1)(2n+1)}{6(n^3 + n^2 + 1)} \dots (v)$$

From Eqs. (i), (ii) and (iv), we get

$$g(n) < f(n) < h(n) \dots (vi)$$

Also, from Eqs (iii) and (v), we have

$$\lim_{n \rightarrow \infty} g(n) = \lim_{n \rightarrow \infty} \frac{\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)}{6\left(1+\frac{1}{n}+\frac{1}{n^2}\right)} = \frac{2}{6} = \frac{1}{3}$$

$$\text{and } \lim_{n \rightarrow \infty} h(n) = \lim_{n \rightarrow \infty} \frac{\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)}{6\left(1+\frac{1}{n}+\frac{1}{n^3}\right)} = \frac{2}{6} = \frac{1}{3}$$

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∴ By applying Sandwich theorem in Eq. (vi), we get

$$\lim_{n \rightarrow \infty} f(n) = \frac{1}{3}$$

Q10 (2) - Single Correct

$$\text{Clearly, } \frac{[x]^2}{x} = \begin{cases} 0, & 0 < x < 1 \\ \frac{1}{x}, & -1 < x < 0 \end{cases}$$

$$\text{and } \frac{[x^2]}{x^2} = \begin{cases} 0, & 0 < x < 1 \\ 0, & -1 < x < 0 \end{cases}$$

Now, consider $\lim_{x \rightarrow 0^+} \frac{[x]^2}{x} = 0$ and $\lim_{x \rightarrow 0^-} \frac{[x]^2}{x} = \lim_{x \rightarrow 0^-} \frac{1}{x}$, which does not exist.

Hence, $\lim_{x \rightarrow 0} \frac{[x]^2}{x} = I$ does not exist.

$$\text{Also, } \lim_{x \rightarrow 0^+} \frac{[x^2]}{x^2} = 0 \text{ and } \lim_{x \rightarrow 0^-} \frac{[x^2]}{x^2} = 0$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{[x^2]}{x^2} = m = 0, \text{ exist}$$

Hence, m exists but I does not.

Q11 (4) - Single Correct

$$g(n+h) = f[\{n+h\}] = f(h) = \lim_{h \rightarrow 0} \frac{e^h - \cos 2h - h}{h^2}$$

$$= \lim_{h \rightarrow 0} \frac{(e^h - 1) + (1 - \cos 2h) - h}{h^2}$$

$$= \lim_{h \rightarrow 0} \left(\frac{e^h - h - 1}{h^2} \right) + \left(\frac{1 - \cos 2h}{h^2} \right)$$

$$= 1/2 + 2 = 5/2$$

$$g(n-h) = f[1 - \{n-h\}] = f(1 - \{-h\})$$

$$f(h) = \lim_{h \rightarrow 0} \frac{e^h - \cos 2h - h}{h^2} = 5/2$$

$$g(n) = 5/2$$

∴ $g(x)$ is continuous for all $x \in I$ and $g(x)$ is continuous for all $x \in R$

Q12 (2) - Single Correct

$$h(x) = \begin{cases} \frac{2 \cos x - \sin 2x}{(\pi - 2x)^2}, & x < \pi/2 \\ \frac{e^{-\cos x} - 1}{8x - 4\pi}, & x > \pi/2 \end{cases}$$

LHL at $x = \pi/2$,

$$\lim_{h \rightarrow 0} \frac{2 \sin h - \sin 2h}{4h^2} = \lim_{h \rightarrow 0} \frac{2 \sin h (1 - \cos h)}{4h^2} = 0$$

RHL at $x = \pi/2$

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$$\lim_{h \rightarrow 0} \frac{e^{\sin h} - 1}{[(\pi/2) + h] - 4\pi} = \lim_{h \rightarrow 0} \frac{e^{\sin h} - 1}{8h} \cdot \frac{\sin h}{\sin h} = \frac{1}{8}$$

$\Rightarrow h(x)$ is discontinuous at $x = \pi/2$.

The irremovable discontinuity at $x = \pi/2$

$$f(\pi^+/2) = 0$$

$$\text{and } g(\pi^-/2) = 1/8$$

$$\Rightarrow f(\pi^+/2) \neq g(\pi^-/2)$$

Q13 (3) - Single Correct

$$f(1^+) = f(1^-) = f(1) = 2$$

$$f(0) = 1, \quad f(2) = 2$$

$$f(2^-) = 1, f(2) = 2$$

$\therefore f$ is not continuous at $x = 2$.

Q14 (2) - Single Correct

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h) + |x|h + xh^2}{h}$$

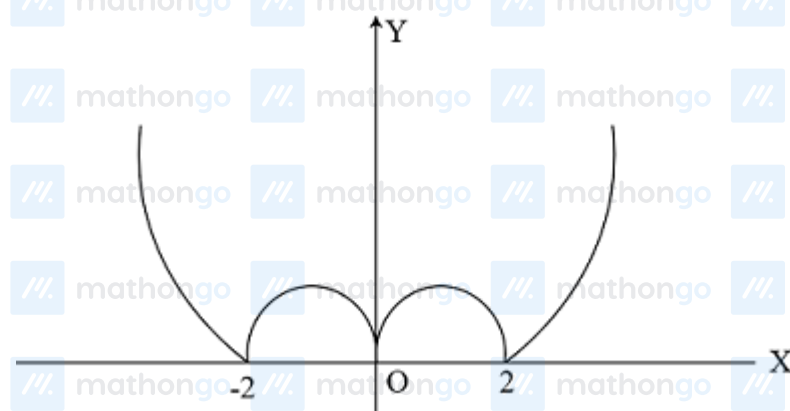
where $x = h$ and $y = x$

$$\therefore f'(x) = \lim_{h \rightarrow 0} \left[\frac{f(h) - f(0)}{h} + |x| + xh \right] \quad [\because f(0) = 0]$$

$$f'(x) = f'(0) + |x| = |x|$$

Q15 (2) - Single Correct

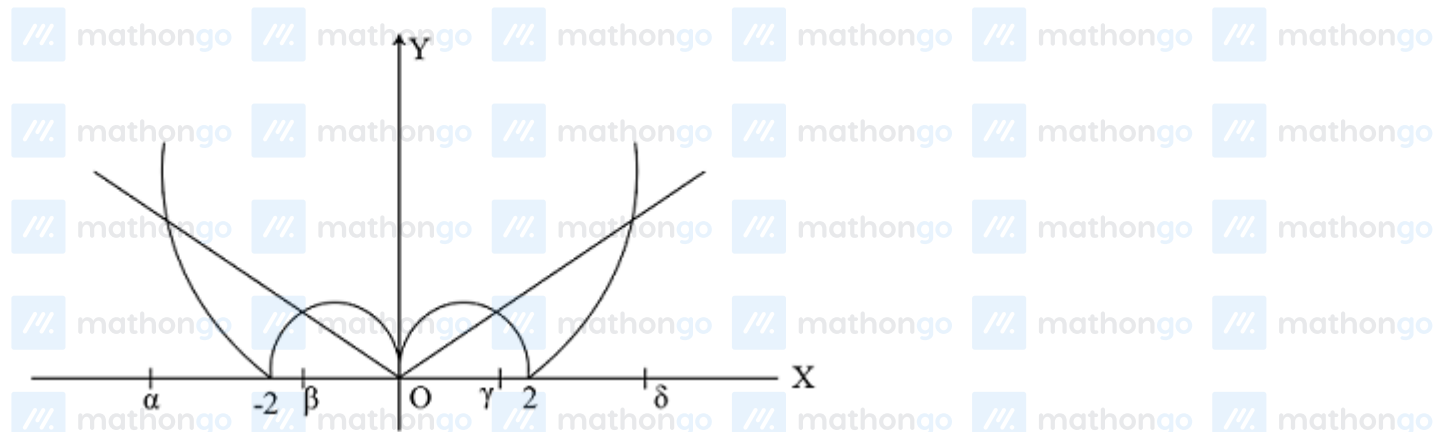
Graph for $y = |x^2 - 2||x|$, is shown as



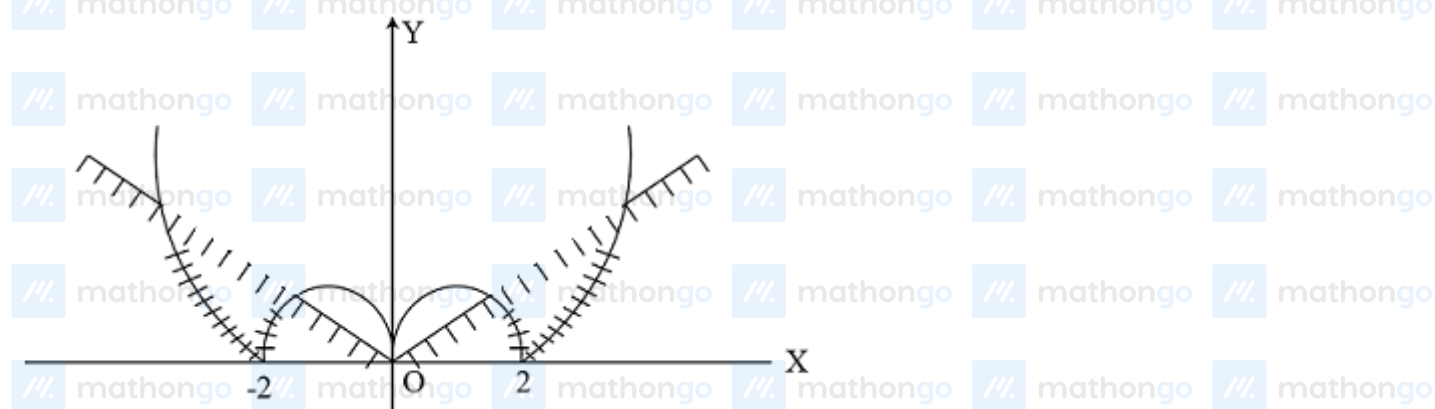
$\therefore |x^2 - 2||x|$ and $|x|$ is shown as

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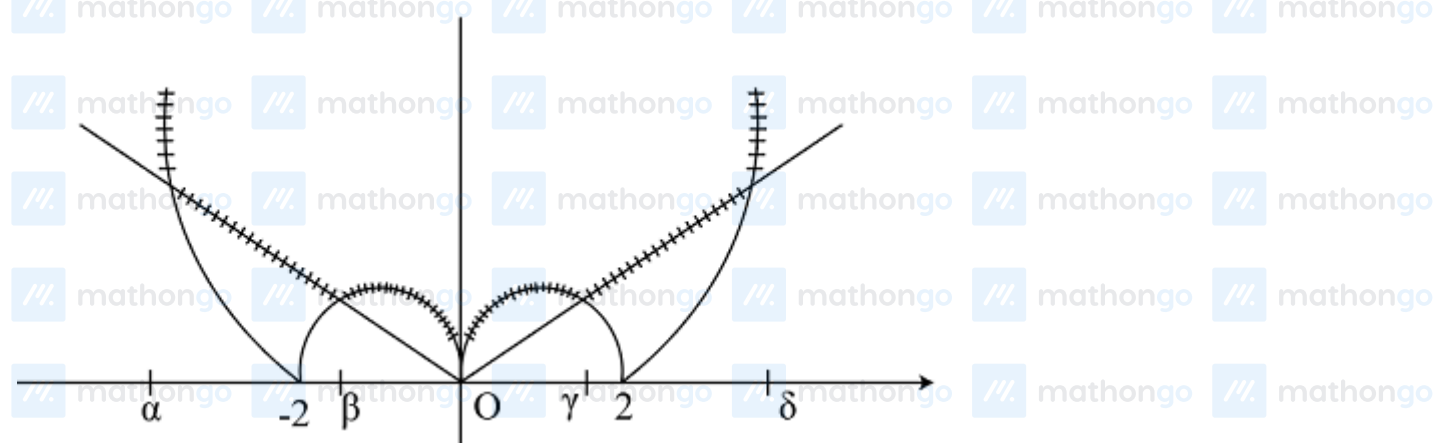


$\therefore f(x) = \max\{|x^2 - 2||x|, |x|\}$ is shown as



$\Rightarrow f(x)$ is not differentiable at 5 points and

$g(x) = \min\{|x^2 - 2||x|, |x|\}$ is shown as



$\Rightarrow g(x)$ is not differentiable at 7 points.

Q16 (1, 3) - Multiple Correct

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$$f(x) = \begin{cases} x^3, & \text{if } x \leq 1 \\ x^2, & \text{if } x > 1 \end{cases}$$

$\Rightarrow f$ is continuous for all x and differentiable for all $x \in (1, \infty)$ Also, $g(x)$ be continuous at $x \in I$

$$g(n) = n^2; g(n^+) = n^2$$

$$g(n^-) = (n-1)^2 + 1$$

$$\therefore n^2 = (n-1)^2$$

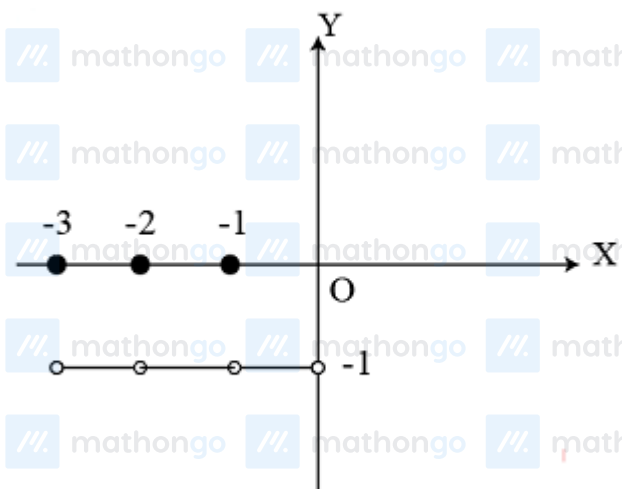
$$\Rightarrow n = 1$$

$g(x)$ is continuous only at one integer $x = 1$

Q17 (1, 2, 3, 4) - Multiple Correct

$$[|x|] - |[x]| = \begin{cases} 0, & \text{if } x = -1 \\ -1, & \text{if } -1 < x < 0 \\ 0, & \text{if } 0 \leq x \leq 1 \\ 0, & \text{if } 1 < x \leq 2 \end{cases}$$

$\therefore$ Range is $\{0, -1\}$



Q18 (2, 3, 4) - Multiple Correct

$$(1) f(x) = \begin{cases} x^2 \sin \frac{1}{x^2}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

is differentiable but $f'(x)$ is not bounded in $[0, 1]$

$$(2) \text{ True, e.g. Let } f(x) = \begin{cases} x, & 0 < x < 1 \\ 1, & x = 0 \\ 0, & x = 1 \end{cases} \text{ bijective but not continuous.}$$

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$$(3) f(x) = x^2 - \cos \pi x + 4$$

$$\text{Now, } f(-1) = 1 - (-1) + 4 = 6$$

$$\text{and } f(2) = 4 - 1 + 4 = 7$$

At f is continuous on $[-1, 2]$, so $f(c) = 2\pi$, for some $c \in (-1, 2)$

$$(4) (f^{-1})'(4) = \frac{1}{f'(0)} = \left(\frac{1}{1+e^x} \right)_{x=0} = \frac{1}{2}$$

Q19 (1, 4) - Multiple Correct

By intermediate value property,

$$\frac{f(0)+f(2)}{2} = f(c), 0 < c < 2$$

By mean value theorem,

$$f(1) - f(0) = f'(c_1), 0 < c_1 < 1$$

$$f(2) - f(1) = f'(c_2), 1 < c_2 < 2$$

By subtracting,

$$f(0) + f(2) - 2f(1) = f'(c_2) - f'(c_1)$$

$$= (c_2 - c_1) f''(c), c_1 < c < c_2$$

$$\Rightarrow f(0) + f(2) - 2f(1) < 0$$

$$\Rightarrow f(0) + f(2) < 2f(1)$$

Q20 (2, 4) - Multiple Correct

$$f'(0^+) = \lim_{h \rightarrow 0} \frac{h^p \cdot \sin \frac{1}{h} + h(\tan h)^q}{h}$$

$$= \lim_{h \rightarrow 0} h^{p-1} \cdot \sin \frac{1}{h} + (\tan h)^q$$

For this limit to exist $(p-1) > 0$ and $q \in \mathbb{N}$.

$$\therefore p \geq 2 \text{ and } q \in \mathbb{N}$$

$$\text{Again, } f'(0^-) = \lim_{h \rightarrow 0} \frac{-h^p \cdot \sin \frac{1}{h} - h(\tanh h)^q}{-h}$$

$$= h^{p-1} \cdot \sin \frac{1}{h} + (\tanh h)^q$$

which is same.

$$\therefore p \geq 2 \text{ and } q \in \mathbb{N}$$

Q21 (1) - Multiple Correct

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$$f(-1) = b(1-1) + 1 = 1$$

$$\text{and } \lim_{h \rightarrow 0} f(-1+h) = 1$$

$$\lim_{h \rightarrow 0} f(-1-h) = \sin[(-1+h+a)\pi]$$

$$= -\sin \pi a$$

$$\text{For continuity, } \sin \pi a = -1 = \sin(2n\pi + 3\pi/2)$$

$$\Rightarrow \pi a = 2n\pi + 3\pi/2$$

$$\Rightarrow a = 2n + 3/2$$

$$\text{Hence, } a = 2n + 3/2, n \in I \text{ and } b \in R.$$

Q22 (1, 2, 3) - Multiple Correct

We have,

$$f(x) = |x+1|(|x| + |x-1|)$$

$$f(x) = \begin{cases} -(x+1)(-x-x+1); & -2 \leq x < -1 \\ (x+1)(-x-x+1); & -1 \leq x < 0 \\ (x+1)(x-x+1); & 0 \leq x < 1 \\ (x+1)(x+x-1); & 1 \leq x < 2 \end{cases}$$

$$= \begin{cases} -(x+1)(1-2x); & -2 \leq x < -1 \\ (x+1)(1-2x); & -1 \leq x < 0 \\ x+1; & 0 \leq x < 1 \end{cases}$$

$$= \begin{cases} (x+1)(2x-1); & 1 \leq x \leq 2 \\ 2x^2+x-1; & -2 \leq x < -1 \\ -2x^2-x+1; & -1 \leq x < 0 \\ x+1; & 0 \leq x < 1 \\ 2x^2+x-1; & 1 \leq x \leq 2 \end{cases}$$

$$\text{Now, } f'(x) = \begin{cases} 4x+1; & -2 \leq x < -1 \\ -4x-1 & -1 \leq x < 0 \\ 1; & 0 \leq x < 1 \\ 4x+1; & 1 \leq x \leq 2 \end{cases}$$

Clearly, $f(x)$ is not differentiable at $x = -1, 0$ and 1 .

Q23 (4) - Multiple Correct

We have, for $-1 < x < 1$

$$\Rightarrow 0 \leq x \sin \pi x \leq \frac{1}{2}$$

Also, $x \sin \pi x$ becomes negative and numerically less than 1, when x is slightly greater than 1 and so by

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definition of $[x]$. $f(x) = [x \sin \pi x] = -1$, when $1 < x < 1 + h$

Thus, $f(x)$ is constant and equal to 0 in the closed interval $[-1, 1]$ and so $f(x)$ is continuous and differentiable in the open interval $(-1, 1)$

At $x = 1$, $f(x)$ is discontinuous, since

$$\lim_{h \rightarrow 0} (1 - h) = 0$$

$$\text{and } \lim_{h \rightarrow 0} (1 + h) = -1$$

$\therefore f(x)$ is not differentiable at $x = 1$

Q24 (1, 4) - Multiple Correct

$$H(x) = \begin{cases} \min\{f(t) : 0 \leq t \leq x\}, & \text{for } 0 \leq x \leq \pi/2 \\ \frac{\pi}{2} - x, & \text{for } \pi/2 < x \leq 3 \end{cases}$$

$$= \begin{cases} \cos x, & \text{for } 0 \leq x \leq \pi/2 \\ \frac{\pi}{2} - x, & \text{for } \pi/2 < x \leq 3 \end{cases}$$

[as, $\min\{f(t) : 0 \leq t \leq x\}$ for $0 \leq x \leq \frac{\pi}{2} = \cos x$, $0 \leq x \leq \frac{\pi}{2}$, as $\cos x$ is decreasing]

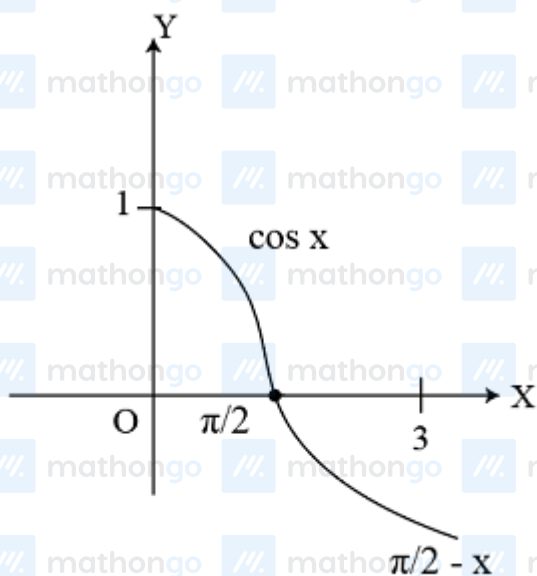
$\therefore H(x)$ is continuous at $x = \frac{\pi}{2}$

$$H'(x) = \begin{cases} -\sin x, & 0 \leq x \leq \pi/2 \\ -1, & \pi/2 < x \leq 3 \end{cases}$$

$$H'\left(\frac{\pi^+}{2}\right) = -1$$

$$H'\left(\frac{\pi^-}{2}\right) = -1$$

$\therefore H(x)$ is differentiable at $x = \frac{\pi}{2}$.



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Also, $H(x)$ is maximum at $x = 0$.

$\Rightarrow$ Maximum value of $H(x) = 1$

Q25 (2, 3) - Multiple Correct

$f(x + y) = f(x) + f(y)$, as $f(x)$ is differentiable at $x = 0$

$\Rightarrow f'(0) = k$

Now, $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$

$\lim_{h \rightarrow 0} \frac{f(x)+f(h)-f(x)}{h}$

$= \lim_{h \rightarrow 0} \lim_{h \rightarrow 0} \frac{f(h)}{h} \left[\frac{0}{0} \text{ form} \right]$

Given, $f(x + y) = f(x) + f(y), \forall x, y$

$\therefore f(0) = f(0) + f(0)$,

when $x = y = 0 \Rightarrow f(0) = 0$

Using L'Hospital's rule,

$= \lim_{h \rightarrow 0} \frac{f(h)}{1} = f'(0) = k \dots (ii)$

$\Rightarrow f(x) = kx$, integrating both sides,

$f(x) = kx + C$, as $f(0) = 0$

$\therefore f(x)$ is continuous for all $x \in R$ and $f'(x) = k$, i.e. constant for all $x \in R$.

Hence, (b) and (c) are correct.

Q26 (3, 4) - Multiple Correct

Given, $f(x)$ is continuous at $x = 0$

(1) $\lim_{x \rightarrow 0} f(x) = 0$

[may or may not be true]

(2) $f(x)$ is continuous at $x = 0$, $\therefore$ It may or may not be continuous at $x = 1$.

(3) $g(x) = x^2 f(x)$ is continuous at $x = 0$, because x^2 and $f(x)$ both are continuous at $x = 0$.

(4) Since, $f(x)$ is continuous at $x = 0$.

$\therefore \lim_{x \rightarrow 0} f(x) = f(0)$

$\Rightarrow \lim_{x \rightarrow 0} [f(x) - f(0)] = 0$

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Q27 (3, 4) - Multiple Correct

Given, $f(x) = \frac{x}{2} - 1$

Since, $f(x)$ is polynomial, it is continuous everywhere.

$f(x)$ is one-one and onto.

$$\therefore y = \frac{x}{2} - 1 \Rightarrow 2y = x - 2$$

$$\Rightarrow x = 2y + 2 \Rightarrow x = 2(y + 1)$$

$$\Rightarrow f^{-1}(x) = 2(x + 1)$$

$f^{-1}(x)$ is also a polynomial function.

It is also continuous.

$$\tan[f(x)] = \tan\left(\frac{x}{2} - 1\right)$$

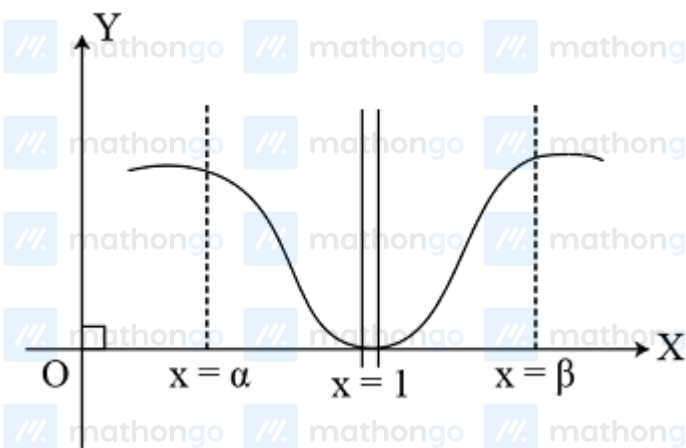
It is continuous in $(0, \pi)$.

$$\text{Now, } \frac{1}{f(x)} = \frac{2}{x-2}$$

it is not continuous at $x = 2$.

Q28 (4) - Paragraph 1

Since, $f(x)$ is symmetric about $x = 1$ and it is twice differentiable. So, $f'(x)$ must have one root at $x = 1$.



$$\therefore f'(x) = a(x - 1)(x - \alpha)(x - \beta)$$

$$f'(x) = a(x - 1)[x^2 - (\alpha + \beta)x + \alpha\beta]$$

$$\text{Here, } \frac{\alpha + \beta}{2} = 1. \text{ So, } \alpha + \beta = 2$$

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$$\begin{aligned}
 f'(x) &= a(x-1)(x^2 - 2x + \alpha\beta) \\
 &= a(x^3 - 2x^2 + \alpha\beta x - x^2 + 2x - \alpha\beta) \\
 &= a(x^3 - 3x^2 + (\alpha\beta + 2)x - \alpha\beta) \\
 f''(x) &= a(3x^2 - 6x + \alpha\beta + 2) \\
 f''(2) &= a(12 - 12 + \alpha\beta + 2) = 0 \\
 \therefore \alpha\beta &= -2 \\
 \therefore \alpha - \frac{2}{\alpha} &= 2 \Rightarrow \alpha^2 - 2\alpha - 2 = 0 \\
 \alpha &= 1 + \sqrt{3}, \beta = 1 - \sqrt{3}
 \end{aligned}$$

Q29 (3) - Paragraph 1

$$\begin{aligned}
 \text{So, sum} &= 1 + \alpha + \beta = 3 \\
 f'(x) &= a(x-1)(x^2 - 2x - 2) \\
 &= a(x-1)[(x-1)^2 - 3] \\
 f(x) &= a \left[\frac{(x-1)^4}{4} - \frac{3}{2}(x-1)^2 \right] + C \\
 \left[\because f(1) = 0, \text{ so } C = 0 \right] \\
 \left[f(2) = 1, \text{ so } a = -4/5 \right] \\
 f(2) &= a[1/4 - 3/2] = 1, \text{ so } a = -4/5 \\
 f(x) &= -4/5 \left[\frac{(x-1)^4}{4} - \frac{3}{2}(x-1)^2 \right] \\
 f(3) &= 8/5 \text{ and } f(1 + \sqrt{3}) = 9/5
 \end{aligned}$$

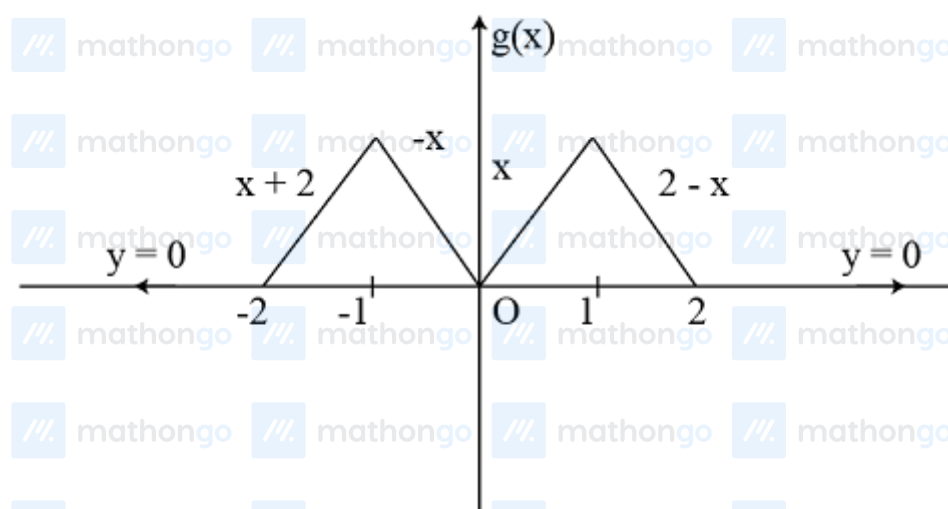
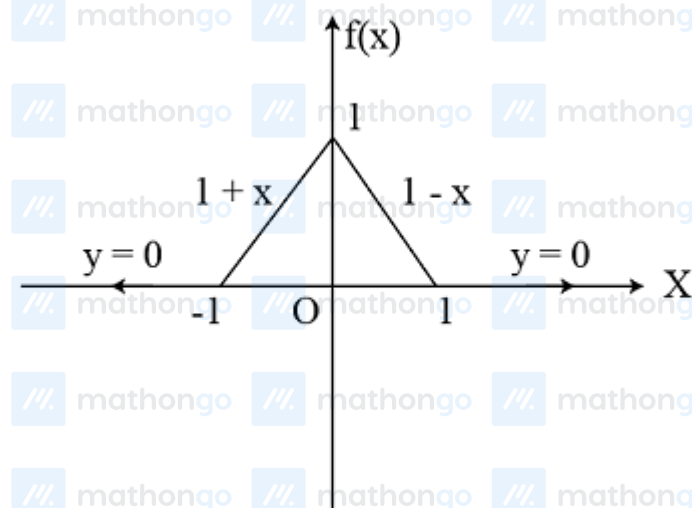
Q30 (2) - Paragraph 2

Explanation (For Question 31, 32)

The graph of $f(x)$ and $g(x)$ intersect at two points $x = \pm 1/2$, $g(x)$ is continuous, $\forall x \in R$ and non-differentiable at $x = -2, -1, 0, 1, 2$ i.e. five points.

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Q31 (1) - Paragraph 2

Q32 (2) - Paragraph 3

Replacing y by $1/x$

$$f(x) \cdot f(1/x) = f(x) + f(1/x)$$

Since $f(1) = 2$, we get

$$f(x) = 1 + x^2$$

Q33 (1) - Paragraph 3

$$f(x) = x^2 + 1$$

$$\Rightarrow f(x) \geq 1$$

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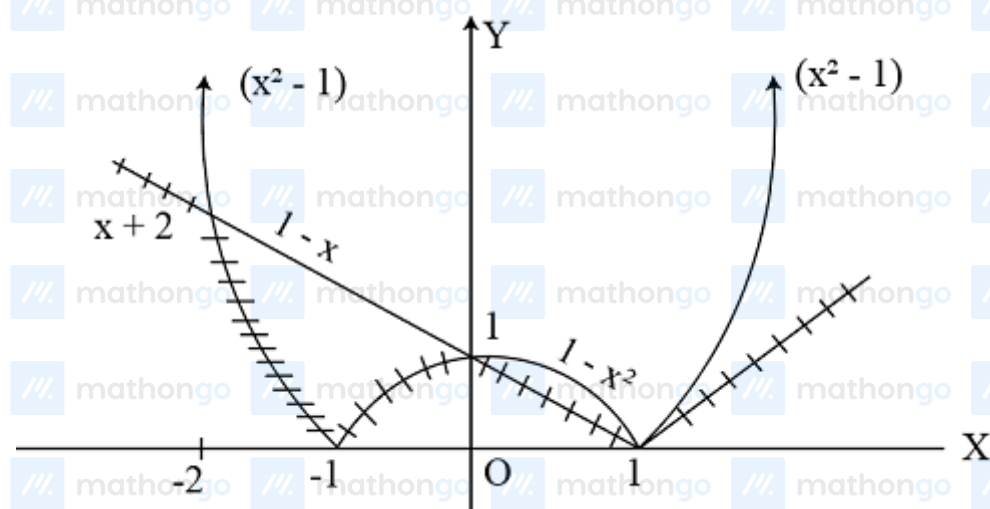
$$0 < \frac{1}{f(x)} \leq 1$$

$$\Rightarrow \left[\frac{1}{f(x)} \right] = \begin{cases} 1, & x = 0 \\ 0, & \text{otherwise} \end{cases}$$

$\therefore g(x)$ has only one point of discontinuity.

Q34 (1) - Paragraph 5

Explanation (For Question 37, 38)



$$f(x) = \begin{cases} x^2 - 1, & x \leq -2 \\ 1 - x, & -2 < x \leq 0 \\ 1 - x^2, & 0 < x \leq 1 \\ x^2 - 1, & x > 1 \end{cases}$$

So, $f(x)$ is continuous everywhere.

Q35 (3) - Paragraph 5

Q36 (1) - Paragraph 6

Explanation (For Question 38, 39)

$$f(x) = \begin{cases} [x], & -2 \leq x \leq -1/2 \\ 2x^2 - 1, & -1/2 < x \leq 2 \end{cases} = \begin{cases} -2, & -2 \leq x < -1 \\ -1, & -1 \leq x \leq -1/2 \\ 2x^2 - 1, & -1/2 < x \leq 2 \end{cases}$$

$$|f(x)| = \begin{cases} 2, & -2 \leq x < -1 \\ 1, & -1 \leq x \leq -1/2 \\ |2x^2 - 1|, & -1/2 < x \leq 2 \end{cases}$$

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$$|f(x)| = \begin{cases} 2, & -2 \leq x < -1 \\ 1, & -1 \leq x \leq -1/2 \\ 1 - 2x^2, & -1/2 < x \leq 1/\sqrt{2} \\ 2x^2 - 1, & 1/\sqrt{2} < x \leq 2 \end{cases}$$

$$f(|x|) = \begin{cases} -2, & -2 \leq |x| < -1 \\ -1, & -1 \leq |x| \leq -1/2 = 2x^2 - 1; -2 \leq x \leq 2 \\ 2|x|^2 - 1, & -1/2 < |x| \leq 2 \end{cases}$$

$$\therefore g(x) = f(|x|) + |f(x)| = \begin{cases} 2x^2 + 1, & -2 \leq x < -1 \\ 2x^2, & -1 \leq x \leq -1/2 \\ 0, & -1/2 < x \leq 1/\sqrt{2} \\ 4x^2 - 2, & 1/\sqrt{2} < x \leq 2 \end{cases}$$

$$\Rightarrow g(-1^-) = \lim_{x \rightarrow -1} (2x^2 + 1) = 3$$

$$\Rightarrow g(-1^+) = \lim_{x \rightarrow -1} (2x^2) = 2$$

$$g\left(-\frac{1}{2}\right) = \lim_{x \rightarrow -1/2} 2x^2 = \frac{1}{2}, g\left(-\frac{1}{2}\right) = 0$$

$$g\left(-\frac{1}{\sqrt{2}}\right) = 0, g\left(-\frac{1}{\sqrt{2}}\right) = \lim_{x \rightarrow -\frac{1}{\sqrt{2}}} (4x^2 - 2) = 0$$

$$\therefore g(x) \text{ is discontinuous at } x = \{-1, -1/2\}.$$

Q37 (4) - Paragraph 6

Q38 (3) - Paragraph 7

Explanation (For Question 41, 42)

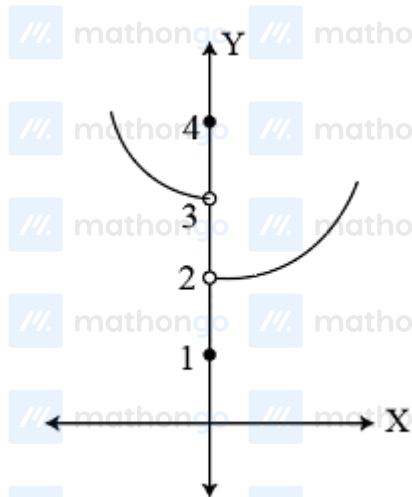
We have, $f(x)$ is exist for all $x \in R - \{0\}$, with $f'(x) < 0, \forall x \in (-\infty, 0)$ and $f'(x) > 0, \forall x \in (0, \infty)$

Also, $\lim_{x \rightarrow 0^-} f(x) = 3; \lim_{x \rightarrow 0^+} f(x) = 2$ and $f(0) = 4$

$\therefore$ Graph of $f(x)$ will be

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$\Rightarrow$ Clearly, as $x \rightarrow 0$, $x^3 - x^2 = x^2(x - 1) \rightarrow 0^-$

and as $x \rightarrow 0$, $2x^4 - x^5 = x^4(2 - x^5) \rightarrow 0^+$

$$\therefore \lim_{x \rightarrow 0} f(x^3 - x^2) = f(0^-) = 3$$

$$\text{and } \lim_{x \rightarrow 0} f(2x^4 - x^5) = f(0^+) = 2$$

Now, consider

$$2 \left[\lim_{x \rightarrow 0} f(x^3 - x^2) \right] = \lambda \left[\lim_{x \rightarrow 0} f(2x^4 - x^5) \right]$$

$$\Rightarrow 2 \times 3 = \lambda(2)$$

$\therefore$ On comparing, $\lambda = 3$

Now, consider,

$$\lim_{x \rightarrow 0^+} \frac{f(-x)x^2}{\left\{ \frac{1 - \cos x}{[f(x)]} \right\}} = \lim_{x \rightarrow 0^+} \frac{f(-x)x^2}{\left(\frac{1 - \cos x}{[f(x)]} \right) - \left[\frac{1 - \cos x}{[f(x)]} \right]} \quad [\because \{x\} = x - [x]]$$

$$= \lim_{x \rightarrow 0^+} \frac{3x^2}{\left(\frac{1 - \cos x}{2} \right) - 0} \quad [\because \lim_{x \rightarrow 0} f(-x) = f(0^-) = 3]$$

$$\lim_{x \rightarrow 0^+} [f(x)] = [f(0^+)] = [2] = 2$$

$$\text{and } x \rightarrow 0^+ \Rightarrow \cos x \rightarrow 1^-$$

$$\therefore [1 - \cos x] = 0$$

$$= 6 \lim_{x \rightarrow 0^+} \frac{x^2}{2 \sin^2 \frac{x}{2}} = 3 \lim_{x \rightarrow 0^+} \frac{4 \times \frac{x^2}{4}}{\sin^2 \frac{x}{2}}$$

$$= 12 \lim_{x \rightarrow 0^+} \frac{\left(\frac{x}{2} \right)^2}{\left[\sin \left(\frac{x}{2} \right) \right]^2} = 12 \times 1 = 12$$

Q39 (2) - Paragraph 7

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Q40 (4) - Integer Type

We have, $a_{n+1} = a_n + \sqrt{1 + a_n^2}$

Let $a_n = \cot(\alpha_n)$

$$\Rightarrow a_{n+1} = \cot(\alpha_n) + \operatorname{cosec}(\alpha_n)$$

$$= \frac{\cos(\alpha_n) + 1}{\sin(\alpha_n)} = \frac{2 \cos^2(\alpha_n/2)}{2 \sin(\alpha_n/2) \cos(\alpha_n/2)} = \cot\left(\frac{\alpha_n}{2}\right) \quad (i)$$

Now, on putting $n = 1$, we get

$$a_1 = \cot(\alpha_1) \text{ and } a_1 = a_0 + \sqrt{1 + a_0^2} = 1$$

$$\Rightarrow \cot(\alpha_1) = 1 \text{ or } \alpha_1 = \frac{\pi}{4}$$

Now, from Eq. (i), we get

$$a_2 = \cot\left(\frac{\alpha_1}{2}\right) = \cot\left(\frac{\pi}{8}\right) = \cot\left(\frac{\pi}{4 \cdot 2}\right)$$

$$a_3 = \cot\left(\frac{\alpha_2}{2}\right) = \cot\left(\frac{\pi}{4 \cdot 2^2}\right)$$

$$a_4 = \cot\left(\frac{\alpha_3}{2}\right) = \cot\left(\frac{\pi}{4 \cdot 2^3}\right)$$

$\vdots$

$$a_n = \cot\left(\frac{\pi}{4 \cdot 2^{n-1}}\right)$$

$$\text{Hence, } \lim_{n \rightarrow \infty} \frac{a_n}{2^{n-1}} = \lim_{n \rightarrow \infty} \frac{\cot\left(\frac{\pi}{4 \cdot 2^{n-1}}\right)}{2^{n-1}}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{\tan\left(\frac{\pi}{4 \cdot 2^{n-1}}\right) 2^{n-1}} = \lim_{n \rightarrow \infty} \frac{\frac{1}{\tan\left(\frac{\pi}{4} \cdot \frac{1}{2^{n-1}}\right)}}{\frac{1}{2^{n-1}} \times \left(\frac{\pi}{4}\right)} \times \left(\frac{\pi}{4}\right)$$

$$= \frac{1}{1 \times \frac{\pi}{4}} = \frac{4}{\pi}$$

Q41 (3) - Integer Type

$$\sin(x_{n+1} - x_n) + 2^{-(n+1)} \cdot \sin x_n \cdot \sin x_{n+1} = 0$$

$$\Rightarrow \cot x_{n+1} - \cot x_n = 2^{-(n+1)}$$

$$\Rightarrow \cot x_n - \cot x_{n-1} = 2^{-n}$$

$\vdots$

$$\Rightarrow \cot x_2 - \cot x_1 = 2^{-2}$$

On adding

$$\cot x_{n+1} - \cot x_1 = \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^{n+1}}$$

$$\Rightarrow \cot x_{n+1} = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^{n+1}}$$

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$$\therefore \lim_{n \rightarrow \infty} \cot x_{n+1} = \frac{1/2}{1-1/2} = 1$$

$$\therefore \lim_{n \rightarrow \infty} x_{n+1} = \pi/4$$

$$\text{or } \lim_{n \rightarrow \infty} x_n = \pi/4 = I$$

$$\therefore 4I = [\pi] = 3$$

Q42 (4) - Integer Type

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{kr}{1 \times 3 \times 5 \times \dots \times (2r+1)} = 1$$

$$\Rightarrow \frac{k}{2} \cdot \lim_{n \rightarrow \infty} \sum_{r=1}^n \left[\frac{1}{1 \times 3 \times 5 \times \dots \times (2r-1)} - \frac{1}{1 \times 3 \times 5 \times \dots \times (2r+1)} \right] = 1$$

$$\Rightarrow \frac{k}{2} \lim_{n \rightarrow \infty} \left[\left(1 - \frac{1}{1 \times 3}\right) + \left(\frac{1}{1 \times 3} - \frac{1}{1 \times 3 \times 5}\right) + \dots + \left(\frac{1}{1 \times 3 \times \dots \times (2n-1)} - \frac{1}{1 \times 3 \times \dots \times (2n+1)}\right) \right] = 1$$

$$\Rightarrow \frac{k}{2} \lim_{n \rightarrow \infty} \left[1 - \frac{1}{1 \times 3 \times 5 \times \dots \times (2n+1)} \right] = 1$$

$$\Rightarrow \frac{k}{2} \times 1 = 1$$

$$\Rightarrow k = 2$$

$$\therefore k^2 = 4$$

Q43 (1) - Integer Type

We have,

$$a_n = n(1 + a_{n-1})$$

$$\Rightarrow a_{n-1} + 1 = \frac{a_n}{n} \dots (i)$$

Now, consider

$$\lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{a_1}\right) \left(1 + \frac{1}{a_2}\right) \dots \left(1 + \frac{1}{a_n}\right) \right] \frac{1}{e}$$

$$= \frac{1}{e} \lim_{n \rightarrow \infty} \frac{(a_1+1)}{a_1} \times \frac{(a_2+1)}{a_2} \times \frac{(a_3+1)}{a_3} \dots \times \frac{(a_n+1)}{a_n}$$

$$= \frac{1}{e} \lim_{n \rightarrow \infty} \left(\frac{a_2}{2} \right) \left(\frac{a_3}{3} \right) \left(\frac{a_4}{4} \right) \dots \left(\frac{a_{n+1}}{n+1} \right) \times \frac{1}{a_1 a_2 a_3 \dots a_n} [\text{using Eq. (i)}]$$

$$[\because a_1 = 1]$$

$$= \frac{1}{e} \lim_{n \rightarrow \infty} \frac{a_{n+1}}{(n+1)!}$$

$$= \frac{1}{e} \lim_{n \rightarrow \infty} \frac{a_{n+1}}{n!} [\text{using Eq. (i)}]$$

$$= \frac{1}{e} \lim_{n \rightarrow \infty} \left(\frac{1}{n!} + \frac{a_n}{n!} \right)$$

$$= \frac{1}{e} \lim_{n \rightarrow \infty} \left(\frac{1}{n!} + \frac{(1+a_{n-1})}{(n-1)!} \right) [\text{using Eq. (i)}]$$

$$= \frac{1}{e} \lim_{n \rightarrow \infty} \left(\frac{1}{n!} + \frac{1}{(n-1)!} + \frac{a_{n-1}}{(n-1)!} \right)$$

$$= \frac{1}{e} \lim_{n \rightarrow \infty} \left(\frac{1}{n!} + \frac{1}{(n-1)!} + \frac{1}{(n-2)!} + \dots + \frac{1}{(2)!} + \frac{1}{1!} + \frac{a_1}{1!} \right)$$

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$$\begin{aligned}
 &= \frac{1}{e} \lim_{n \rightarrow \infty} \left(1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} \right) \\
 &= \frac{1}{e} \left(1 + \frac{1}{1!} + \frac{1}{2!} + \cdots \infty \right) \\
 &= \frac{1}{e} \cdot e = 1 \quad \left[\because e = 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots \infty \right]
 \end{aligned}$$

Q44 (4) - Integer Type

$$\therefore [x] + [x + 1/4] + [x + 2/4] + [x + 3/4] = [4x]$$

$\therefore f(x) = [4x]$, which becomes discontinuous at

$$x = 1/4, 2/4, 3/4, 1 \in [0, 1]$$

Q45 (5) - Integer Type

$g(x)$ must assume only one integer value

$$\text{i.e. } g(x) = 0, \quad \forall x \in R$$

$$\text{where } \frac{2}{c} < \frac{f(x)}{c} < \frac{\sqrt{21}}{c}$$

$$\Rightarrow \frac{\sqrt{21}}{c} < 1 \Rightarrow c > \sqrt{21}$$

$\therefore$ Least positive integer is 5

Q46 (2) - Integer Type

$$\text{When, } 0 < x < 2 \Rightarrow 0 < \frac{x}{2} < 1$$

$$\therefore 0 < \frac{\pi x}{2} < \frac{\pi}{2} \Rightarrow 0 < \left(\cos \frac{\pi x}{2} \right) < 1$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left(\cos \frac{\pi x}{2} \right)^{2n} \rightarrow 0$$

and $f(2) = 1$

$\therefore f(x)$ is discontinuous at $x = 2$.

Q47 (1) - Integer Type

$$\text{Here, } f'(x) = 3x^2 - 2x + 1 > 0, \forall x \in R$$

$$\Rightarrow \max f(t) = f(x), \text{ when } 0 \leq t \leq x$$

$$g(x) = \begin{cases} x^3 - x^2 + x + 1, & 0 \leq x \leq 1 \\ 3 - x, & 1 < x \leq 2 \end{cases}$$

$\Rightarrow g(x)$ is continuous but not differentiable at $x = 1$

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Q48 (6) - Integer Type

$$\text{Given, } f(x) = ([2x] + \{2x\})([2x] - \{2x\}) = 4x - 4x\{2x\}$$

$$2x \in (-4, 4)$$

Hence $f(x)$ is discontinuous when $2x = -3, -2, -1, 1, 2, 3$.

At $x = 0$, $f(x)$ is continuous

Q49 (0) - Integer Type

Replacing x by $-x$

$$f(2x^2 - 1) = -2x^3 f(-x)$$

$$\therefore 2x^3 f(x) = -2x^3 f(-x)$$

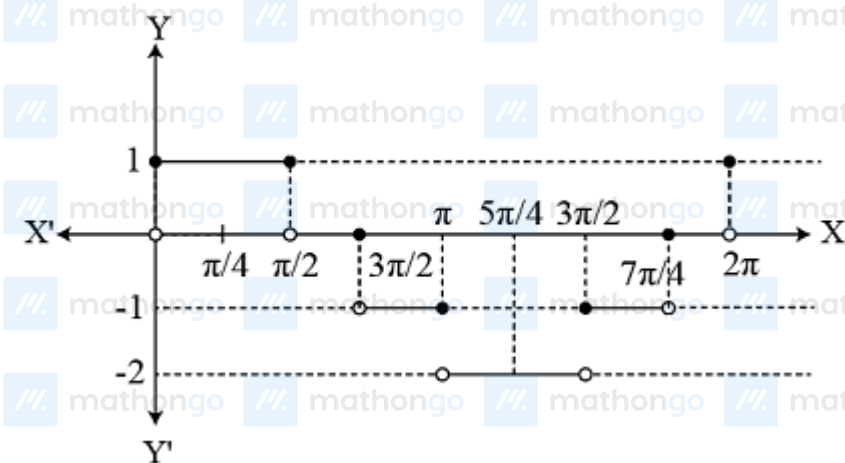
$$\Rightarrow f(-x) = -f(x)$$

$$\therefore f^{2016}(0) = 0$$

Q50 (5) - Integer Type

Given, $f(x) = [\sin x + \cos x] = [g(x)]$, where

$$g(x) = \sin x + \cos x$$



Clearly, from the graph $f(x)$ is discontinuous at

$$x = 0, \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{3\pi}{2}, \frac{7\pi}{4}, 2\pi$$

$$\therefore x \in \left\{ \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{3\pi}{2}, \frac{7\pi}{4} \right\}$$